

Project Report

D.A.M.S – Digital Asset Management System

Abstract

D.A.M.S (Digital Asset Management System) is a secure web-based digital vault designed to protect digital assets through encryption, auditing, and controlled access. The platform combines AES-based encryption, blockchain-inspired audit logging, and a modern web interface to provide secure storage and monitoring of sensitive digital content.

Introduction

Traditional storage platforms often depend on server-side trust and provide limited transparency for security events. D.A.M.S addresses this challenge by integrating encryption, access management, audit monitoring, and secure storage into a unified experience.

Objectives

- Provide secure digital asset storage
- Encrypt uploaded assets before storage
- Maintain tamper-aware audit records
- Deliver an intuitive user experience
- Support secure authentication and session handling
- Enable scalable cloud deployment

Technologies Used

Frontend: React + Vite + TypeScript

Backend: Node.js

Database: MongoDB

Security: AES Encryption, SHA-based audit verification

UI: Tailwind CSS

Methodology

1. User authentication and session validation
2. Secure asset upload
3. Encryption before storage
4. Audit event generation
5. Asset retrieval and decryption
6. Monitoring and access management

Implementation Modules

- Authentication Module
- Secure Vault Module
- Encryption Engine
- Audit & Monitoring Hub

- Asset Explorer
- Dashboard and Analytics

Results

- Functional secure vault implementation
- Encrypted asset lifecycle
- Improved audit visibility
- Modern responsive interface

Conclusion

D.A.M.S demonstrates that advanced security practices can be integrated into a premium web experience while maintaining usability and performance.

Future Scope

- Multi-cloud storage
- Hardware MFA integration
- Advanced anomaly detection
- Enterprise access policies